

The Limit Laws

If both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, and " c " is a constant, then

$$1) \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$2) \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$3) \lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$4) \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \quad \lim_{x \rightarrow a} g(x) \neq 0$$

$$5) \lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$$

$$6) \lim_{x \rightarrow a} c = c$$

$$7) \lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) \text{ if "f" is continuous where its input is } \lim_{x \rightarrow a} g(x).$$

8) If " f " is continuous at $x=a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.
*Remember, this is the definition of continuity at $x=a$.

Implicit Differentiation

A Proof of the Rule for Exponential Functions

$$iX) \frac{d}{dx}(a^x) = a^x \ln a$$

If $f(x) = a^x$, where "a" is a constant and $a > 0$,

$$\text{then } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a^{x+h} + a^x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a^x a^h + a^x}{h} \quad \text{Exponent Rule \#1}$$

$$= \lim_{h \rightarrow 0} \frac{a^x (a^h + 1)}{h}$$

$$= \lim_{h \rightarrow 0} \left[a^x \left(\frac{a^h + 1}{h} \right) \right]$$

$$\stackrel{3}{=} \lim_{h \rightarrow 0} a^x \lim_{h \rightarrow 0} \frac{a^h + 1}{h}$$

$$\stackrel{6}{=} a^x \ln a$$

We find with a table or graph that

Differentiation Formulas

Constant Rule i) $\frac{d}{dx}(c) = 0$ "c" is a constant.

Power Rule ii) $\frac{d}{dx}(x^n) = nx^{n-1}$ "n" is a constant.

Constant Multiple Rule iii) $\frac{d}{dx}[cf(x)] = cf'(x)$ "c" is a constant.

Sum Rule iv) $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$

Difference Rule v) $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$

Product Rule vi) $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

Quotient Rule vii) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

The Chain Rule viii) If $h(x) = f(g(x))$, then

$$h'(x) = f'(g(x))g'(x)$$

Rule for Exponential Functions iX) $\frac{d}{dx}(a^x) = a^x \ln a$, "a" is a constant and $a > 0$.

Rule for base "e"
Exponential Functions

$$x) \frac{d}{dx}(e^x) = e^x$$

Rule for Logarithmic Functions

$$xi) \frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}, \text{ "a" is a constant and } a > 0.$$

Rule for the Natural Logarithm Function

$$xii) \frac{d}{dx}(\ln x) = \frac{1}{x}$$

Rule for Composite Natural Logarithm Functions

$$xiii) \frac{d}{dx} \{ \ln[g(x)] \} = \frac{g'(x)}{g(x)}$$

Trigonometric Differentiation Formulas

$$I) \frac{d}{dx}(\sin x) = \cos x$$

$$II) \frac{d}{dx}(\cos x) = -\sin x$$

$$III) \frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$IV) \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$V) \frac{d}{dx}(\tan x) = \sec^2 x$$

$$VI) \frac{d}{dx}(\cot x) = -\csc^2 x$$

Differentiating Both Sides of an Equation

$$\text{If } A = B$$

$$\text{then } A' = B'$$

Explicit Differentiation: Finding the derivative after the dependent variable (usually "y") has been isolated or "solved for."

Examples:

$$y = x^2$$

$$y' = 2x$$

$$y = \sin x + \frac{1}{x^2}$$

$$y' = \cos x - \frac{1}{x^3}$$

$$y = 3\sqrt{x} + 4$$

$$y' = \frac{3}{2\sqrt{x}}$$

Implicit Differentiation: Finding the derivative when the dependent variable (usually "y") has not been isolated or "solved for." In this scenario, the chain rule is often necessary.

Example: $x^4 + 2x^3 + y^5 = 2y - x - 3y^2 + \tan y + \tan x$

$$4x^3 + 6x^2 + 5y^4 \cdot y' = 2y' - 1 - 3(2y \cdot y') + y' \sec^2 y + \sec^2 x$$

Chain rule

$$5y^4 y' + 6y y' - y' \sec^2 y - 2y' = -4x^3 - 6x^2 - 1 + \sec^2 x$$

$$y'(5y^4 + 6y - \sec^2 y - 2) = -4x^3 - 6x^2 - 1 + \sec^2 x$$

$$y' = \frac{-4x^3 - 6x^2 - 1 + \sec^2 x}{5y^4 + 6y - \sec^2 y - 2}$$

Find y' using implicit differentiation.

• ① $\cos x - 1 = y + y^4 + x^3$

$$-\sin x + 0 = y' + 4y^3 \cdot y' + 3x^2$$

$$-\sin x - 3x^2 = y' + 4y^3 \cdot y'$$

$$-\sin x - 3x^2 = y'(1 + 4y^3)$$

$$\frac{-\sin x - 3x^2}{1 + 4y^3} = \frac{y'(1 + 4y^3)}{1 + 4y^3}$$

$$\boxed{\frac{-\sin x - 3x^2}{1 + 4y^3}} = y'$$

② $5y^3 + y^2 - \sin y = 2x^6 - 6y$

$$5(3y^2 \cdot y') + 2y^{2-1} \cdot y' - y' \cos y = 12x^5 - 6y'$$

$$15y^2 \cdot y' + 2y \cdot y' - y' \cos y + 6y' = 12x^5$$

$$y'(15y^2 + 2y - \cos y + 6) = 12x^5$$

$$\frac{y'(15y^2 + 2y - \cos y + 6)}{15y^2 + 2y - \cos y + 6} = \frac{12x^5}{15y^2 + 2y - \cos y + 6}$$

$$y' = \boxed{\frac{12x^5}{15y^2 + 2y - \cos y + 6}}$$

$$\textcircled{3} \quad y - \frac{y^5}{2} + \sec x = 4x - y^{10} - \csc y$$

$$y - \frac{1}{2}y^5 + \sec x = 4x - y^{10} - \csc y$$

$$y' - \frac{1}{2}(5y^4 \cdot y') + \sec x \tan x = 4 - 10y^9 \cdot y' - (-\csc y \cot y) \cdot y'$$

$$y' - \frac{5}{2}y^4 \cdot y' + \sec x \tan x = 4 - 10y^9 \cdot y' + y' \csc y \cot y$$

$$y' - \frac{5}{2}y^4 \cdot y' + 10y^9 y' - y' \csc y \cot y = 4 - \sec x \tan x$$

$$y' (1 - \frac{5}{2}y^4 + 10y^9 - \csc y \cot y) = 4 - \sec x \tan x$$

$$\frac{y' (1 - \frac{5}{2}y^4 + 10y^9 - \csc y \cot y)}{1 - \frac{5}{2}y^4 + 10y^9 - \csc y \cot y} = \frac{4 - \sec x \tan x}{1 - \frac{5}{2}y^4 + 10y^9 - \csc y \cot y}$$

$$y' = \boxed{\frac{4 - \sec x \tan x}{1 - \frac{5}{2}y^4 + 10y^9 - \csc y \cot y}}$$

$$\textcircled{4} \quad -3x^2 + 4y = 2y^2 + x$$

$$-6x + 4y' = 2(2y^{2-1} \cdot y') + 1$$

$$-6x + 4y' = 4y \cdot y' + 1$$

$$4y' - 4y \cdot y' = 6x + 1$$

$$y'(4 - 4y) = 6x + 1$$

$$\frac{y'(4 - 4y)}{4 - 4y} = \frac{6x + 1}{4 - 4y}$$

$$y' = \boxed{\frac{6x + 1}{4 - 4y}}$$

• ⑤ $\cos y - 2x^3 = y^8$

$$-y' \sin y - 6x^2 = 8y^7 y'$$

$$-6x^2 = 8y^7 y' + y' \sin y$$

$$-6x^2 = y' (8y^7 + \sin y)$$

$$\frac{-6x^2}{8y^7 + \sin y} = \frac{y' (8y^7 + \sin y)}{8y^7 + \sin y}$$

$$\boxed{\frac{-6x^2}{8y^7 + \sin y}} = y'$$

⑥ $-3y + 2 - 3x = 10y^2 + x^2$

$$-3y' + 0 - 3 = 10(2y^{2-1} y') + 2x$$

$$-3y' - 3 = 20y \cdot y' + 2x$$

$$-3 - 2x = 20y \cdot y' + 3y'$$

$$-3 - 2x = y' (20y + 3)$$

$$\frac{-3 - 2x}{20y + 3} = \frac{y' (20y + 3)}{20y + 3}$$

$$\boxed{\frac{-2x - 3}{20y + 3}} = y'$$

$$\textcircled{7} \pi - y = \csc x + \cot y$$

$$0 - y' = -\csc x \cot x - y' \csc^2 y$$

$$-y' = -\csc x \cot x - y' \csc^2 y$$

$$y' \csc^2 y - y' = -\csc x \cot x$$

$$y' (\csc^2 y - 1) = -\csc x \cot x$$

$$\frac{y' (\csc^2 y - 1)}{\csc^2 y - 1} = \frac{-\csc x \cot x}{\csc^2 y - 1}$$

$$y' = \boxed{-\frac{\csc x \cot x}{\csc^2 y - 1}}$$

$$\textcircled{8} 4x^2 + 2x - 1 = 4y^3 + y^2$$

$$8x + 2 + 0 = 4(3y^2 \cdot y') + 2y^{2-1} \cdot y'$$

$$8x + 2 = 12y^2 \cdot y' + 2y \cdot y'$$

$$8x + 2 = y' (12y^2 + 2y)$$

$$\frac{8x + 2}{12y^2 + 2y} = \frac{y' (12y^2 + 2y)}{12y^2 + 2y}$$

$$\frac{8x + 2}{12y^2 + 2y} = y'$$

$$\boxed{\frac{4x + 1}{6y^2 + y}} = y'$$

• ⑨ $3^{\cos x} - 8^x + 2x = 4^y + 10y + e^x - e^{x^3+10} + e^y$

$$(3^{\cos x} \ln 3)(-\sin x) - 8^x \ln 8 + 2 = 4^y (\ln 4) y' + 10y' + e^x - e^{x^3+10} (3x^2) + e^y y'$$

$$-3^{\cos x} (\ln 3) \sin x - 8^x \ln 8 + 2 - e^x + 3x^2 e^{x^3+10} = 4^y (\ln 4) y' + 10y' + y' e^y$$

$$-3^{\cos x} (\ln 3) \sin x - 8^x \ln 8 + 2 - e^x + 3x^2 e^{x^3+10} = y' (4^y \ln 4 + 10 + e^y)$$

$$\frac{-3^{\cos x} (\ln 3) \sin x - 8^x \ln 8 + 2 - e^x + 3x^2 e^{x^3+10}}{4^y \ln 4 + 10 + e^y} = y'$$

$$\boxed{\frac{-3^{\cos x} (\ln 3) \sin x - 8^x \ln 8 + 2 - e^x + 3x^2 e^{x^3+10}}{4^y \ln 4 + 10 + e^y}} = y'$$

• ⑩ $e^x + 2^y - 3x^5 = 6 + e^y + 4y^4 + 7^x + e^{\sin x} - 5^{4-7x}$

$$e^x + 2^y (\ln 2) y' - 15x^4 = 0 + e^y y' + 4(4y^3 y') + 7^x \ln 7 + e^{\sin x} \cos x - 5^{4-7x} (\ln 5) (-7)$$

$$2^y (\ln 2) y' - e^y y' - 16y^3 y' = -e^x + 15x^4 + 7^x \ln 7 + e^{\sin x} \cos x + 7(5^{4-7x} \ln 5)$$

$$y' (2^y \ln 2 - e^y - 16y^3) = -e^x + 15x^4 + 7^x \ln 7 + e^{\sin x} \cos x + 7(5^{4-7x} \ln 5)$$

$$\frac{y' (2^y \ln 2 - e^y - 16y^3)}{2^y \ln 2 - e^y - 16y^3} = \frac{-e^x + 15x^4 + 7^x \ln 7 + e^{\sin x} \cos x + 7(5^{4-7x} \ln 5)}{2^y \ln 2 - e^y - 16y^3}$$

$$y' = \boxed{\frac{-e^x + 15x^4 + 7^x \ln 7 + e^{\sin x} \cos x + 7(5^{4-7x} \ln 5)}{2^y \ln 2 - e^y - 16y^3}}$$

$$xi) \frac{d}{dx} (\log_a x) = \frac{1}{x \ln a}$$

A Proof of the Rule for Logarithmic Functions

If $y = \log_a x$, where "a" is a constant and $a > 0$, then by the definition of logarithms,

$$a^y = x$$

Use implicit differentiation

$$\frac{d}{dx} (a^y) = \frac{d}{dx} (x)$$

$$a^y (\ln a) \cdot y' = 1$$

The chain rule

Isolate y'

$$\frac{a^y (\ln a) y'}{a^y \ln a} = \frac{1}{a^y \ln a}$$

$$y' = \frac{1}{a^y \ln a}$$

$$y' = \frac{1}{a^{\log_a x} \ln a}$$

Logarithm Property IV

$$y' = \frac{1}{x \ln a}$$

Find y' using implicit differentiation.

$$\textcircled{II} \ln x - \log_5 x + \log_3 y = 2x + \ln y + \log_8 (2x+1) + \ln (x^7 - 2x)$$

$$\frac{1}{x} - \frac{1}{x \ln 5} + \frac{1}{y \ln 3} \cdot y' = 2 + \frac{y'}{y} + \frac{1}{(2x+1) \ln 8} \cdot 2 + \frac{7x^6 - 2}{x^7 - 2x}$$

$$\frac{1}{x} - \frac{1}{x \ln 5} - 2 - \frac{2}{(2x+1) \ln 8} - \frac{7x^6 - 2}{x^7 - 2x} = \frac{y'}{y} - \frac{1}{y \ln 3} y'$$

$$\frac{1}{x} - \frac{1}{x \ln 5} - 2 - \frac{2}{(2x+1) \ln 8} - \frac{7x^6 - 2}{x^7 - 2x} = \left(\frac{1}{y} - \frac{1}{y \ln 3} \right) y'$$

$$\frac{\frac{1}{x} - \frac{1}{x \ln 5} - 2 - \frac{2}{(2x+1) \ln 8} - \frac{7x^6 - 2}{x^7 - 2x}}{\frac{1}{y} - \frac{1}{y \ln 3}} = \frac{\left(\frac{1}{y} - \frac{1}{y \ln 3} \right) y'}{\frac{1}{y} - \frac{1}{y \ln 3}}$$

$$\boxed{\frac{\frac{1}{x} - \frac{1}{x \ln 5} - 2 - \frac{2}{(2x+1) \ln 8} - \frac{7x^6 - 2}{x^7 - 2x}}{\frac{1}{y} - \frac{1}{y \ln 3}}} = y'$$

Logarithm Property I

Logarithmic equations and exponential equations are equivalent. In other words, $x = \log_b C$ if and only if $C = b^x$. We learned in geometry class that "if and only if" means the converse of the statement is true also. Note: Traditionally, neither "C" nor "b" is allowed to be negative or zero when placed in a logarithmic expression.

Logarithm Property II

$$\log_b b^x = x$$

Logarithm Property III

$$\log_b b = 1$$

Logarithm Property IV

$$b^{\log_b X} = X$$

Logarithm Property V

$$\log_k(AB) = \log_k A + \log_k B$$

Logarithm Property VI

$$\log_k\left(\frac{A}{B}\right) = \log_k A - \log_k B$$

Logarithm Property VII

$$\log_k A^B = B \log_k A$$

Logarithm Property VIII

$$\text{If } \log_k A = \log_k B, \text{ then } A = B.$$

• ⑫ $\log_2 x - \log_4(x^2) + \ln x + x^7 + 4y^3 = 3 - \log_5 y + \ln y - \ln(\csc x)$

$$\frac{1}{x \ln 2} - \frac{1}{x^2 \ln 4} (2x) + \frac{1}{x} + 7x^6 + 4(3y^2 y') = 0 + \frac{1}{y \ln 5} y' - \left(\frac{-\csc x \cot x}{\csc x} \right)$$

$$\frac{1}{x \ln 2} - \frac{2}{x \ln 4} + \frac{1}{x} + 7x^6 + 12y^2 y' = \frac{y'}{y \ln 5} + \cot x$$

$$\frac{1}{x \ln 2} - \frac{2}{x \ln 4} + \frac{1}{x} + 7x^6 - \cot x = \frac{y'}{y \ln 5} - 12y^2 y'$$

$$\frac{1}{x \ln 2} - \frac{2}{x \ln 4} + \frac{1}{x} + 7x^6 - \cot x = y' \left(\frac{1}{y \ln 5} - 12y^2 \right)$$

$$\frac{\frac{1}{x \ln 2} - \frac{2}{x \ln 4} + \frac{1}{x} + 7x^6 - \cot x}{\frac{1}{y \ln 5} - 12y^2} = \frac{y' \left(\frac{1}{y \ln 5} - 12y^2 \right)}{\frac{1}{y \ln 5} - 12y^2}$$

$$\boxed{\frac{\frac{1}{x \ln 2} - \frac{2}{x \ln 4} + \frac{1}{x} + 7x^6 - \cot x}{\frac{1}{y \ln 5} - 12y^2}} = y'$$

• ⑬ $xy + x^3 = 3y^7$

$$\left[\frac{d}{dx}(x) \right] y + xy' + 3x^2 = 3(7y^6 y')$$

$$(1) y + xy' + 3x^2 = 21y^6 y'$$

$$y + 3x^2 = 21y^6 y' - xy'$$

$$y + 3x^2 = y'(21y^6 - x)$$

$$\frac{y + 3x^2}{21y^6 - x} = \frac{y'(21y^6 - x)}{21y^6 - x}$$

$$\boxed{\frac{y + 3x^2}{21y^6 - x}} = y'$$

$$(14) \quad 4y + 3x = 10 - 3y^5 \cot x$$

$$4y' + 3 = 0 + \left[\frac{d}{dx} (-3y^5) \right] \cot x + (-3y^5) \frac{d}{dx} (\cot x)$$

$$4y' + 3 = -3(5y^4 \cdot y') \cot x - 3y^5 (-\csc^2 x)$$

$$4y' + 3 = -15y^4 y' \cot x + 3y^5 \csc^2 x$$

$$4y' + 15y^4 y' \cot x = 3y^5 \csc^2 x - 3$$

$$y' (4 + 15y^4 \cot x) = 3y^5 \csc^2 x - 3$$

$$\frac{y' (4 + 15y^4 \cot x)}{4 + 15y^4 \cot x} = \frac{3y^5 \csc^2 x - 3}{4 + 15y^4 \cot x}$$

$$y' = \boxed{\frac{3y^5 \csc^2 x - 3}{4 + 15y^4 \cot x}}$$

$$(15) \quad 4x^{10} + y^3 = y^2 x^3$$

$$40x^9 + 3y^2 y' = \left[\frac{d}{dx} (y^2) \right] x^3 + y^2 \frac{d}{dx} (x^3)$$

$$40x^9 + 3y^2 y' = 2y^{2-1} y' x^3 + y^2 (3x^2)$$

$$40x^9 + 3y^2 y' = 2y y' x^3 + 3y^2 x^2$$

$$40x^9 - 3y^2 x^2 = 2y y' x^3 - 3y^2 y'$$

$$40x^9 - 3y^2 x^2 = y' (2yx^3 - 3y^2)$$

$$\frac{40x^9 - 3y^2 x^2}{2yx^3 - 3y^2} = \frac{y' (2yx^3 - 3y^2)}{2yx^3 - 3y^2}$$

$$\boxed{\frac{40x^9 - 3y^2 x^2}{2yx^3 - 3y^2}} = y'$$

$$(16) \quad x \sin y = 7x - y$$

$$\left[\frac{d}{dx}(x) \right] \sin y + x \frac{d}{dx}(\sin y) = 7 - y'$$

$$(1) \sin y + x y' \cos y = 7 - y'$$

$$x y' \cos y + y' = 7 - \sin y$$

$$y' (x \cos y + 1) = 7 - \sin y$$

$$\frac{y' (x \cos y + 1)}{x \cos y + 1} = \frac{7 - \sin y}{x \cos y + 1}$$

$$y' = \boxed{\frac{7 - \sin y}{x \cos y + 1}}$$

$$(17) \quad \frac{x}{y+1} = 4x + y^2$$

$$\frac{\left[\frac{d}{dx}(x) \right] (y+1) - x \frac{d}{dx}(y+1)}{(y+1)^2} = 4 + 2y y'$$

$$\frac{(1)(y+1) - x y'}{(y+1)^2} = 4 + 2y y'$$

$$(y+1)^2 \frac{y+1 - x y'}{(y+1)^2} = (4 + 2y y') (y+1)^2$$

$$y+1 - x y' = 4(y+1)^2 + 2y y' (y+1)^2$$

$$y+1 - 4(y+1)^2 = 2y y' (y+1)^2 + x y'$$

$$y+1 - 4(y+1)^2 = y' [2y(y+1)^2 + x]$$

$$\frac{y+1 - 4(y+1)^2}{2y(y+1)^2 + x} = \frac{y' [2y(y+1)^2 + x]}{2y(y+1)^2 + x}$$

$$\boxed{\frac{y+1 - 4(y+1)^2}{2y(y+1)^2 + x}} = y'$$

$$(18) x^3 + 2 = \frac{2y^4}{3y^5 + x}$$

$$3x^2 + 0 = \frac{\left[\frac{d}{dx}(2y^4)\right](3y^5 + x) - 2y^4 \frac{d}{dx}(3y^5 + x)}{(3y^5 + x)^2}$$

$$3x^2 = \frac{2(4y^3 y')(3y^5 + x) - 2y^4(3(5y^4 y') + 1)}{(3y^5 + x)^2}$$

$$(3y^5 + x)^2 3x^2 = \frac{8y^3 y'(3y^5 + x) - 2y^4(15y^4 y' + 1)}{(3y^5 + x)^2} (3y^5 + x)^2$$

$$3x^2(3y^5 + x)^2 = 24y^8 y' + 8y^3 y' x - 30y^8 y' - 2y^4$$

$$3x^2(3y^5 + x)^2 + 2y^4 = 24y^8 y' + 8y^3 y' x - 30y^8 y'$$

$$3x^2(3y^5 + x)^2 + 2y^4 = y'(24y^8 + 8y^3 x - 30y^8)$$

$$\frac{3x^2(3y^5 + x)^2 + 2y^4}{-6y^8 + 8y^3 x} = \frac{y'(-6y^8 + 8y^3 x)}{-6y^8 + 8y^3 x}$$

$$\boxed{\frac{3x^2(3y^5 + x)^2 + 2y^4}{-6y^8 + 8y^3 x}} = y'$$

$$(19) \cos x = \frac{y^2}{3x+2} - y$$

$$-\sin x = \frac{\left[\frac{d}{dx}(y^2)\right](3x+2) - y^2 \frac{d}{dx}(3x+2)}{(3x+2)^2} - y'$$

$$-\sin x = \frac{2y y'(3x+2) - y^2(3)}{(3x+2)^2} - y'$$

$$y' - \sin x = \frac{6y y' x + 4y y' - 3y^2}{(3x+2)^2}$$

$$(3x+2)^2 (y' - \sin x) = \frac{6y y' x + 4y y' - 3y^2}{(3x+2)^2} (3x+2)^2$$

$$y'(3x+2)^2 - (3x+2)^2 \sin x = 6y y' x + 4y y' - 3y^2$$

$$3y^2 - (3x+2)^2 \sin x = 6y y' x + 4y y' - y'(3x+2)^2$$

$$3y^2 - (3x+2)^2 \sin x = y' [6yx + 4y - (3x+2)^2]$$

$$\frac{3y^2 - (3x+2)^2 \sin x}{6yx + 4y - (3x+2)^2} = \frac{y' [6yx + 4y - (3x+2)^2]}{6yx + 4y - (3x+2)^2}$$

$$\boxed{\frac{3y^2 - (3x+2)^2 \sin x}{6yx + 4y - (3x+2)^2}} = y'$$

$$\textcircled{20} \frac{\sin y}{1+x} = 2y^8 - 3$$

$$\frac{\left[\frac{d}{dx}(\sin y) \right] (1+x) - \sin y \frac{d}{dx}(1+x)}{(1+x)^2} = 2(8y^7 \cdot y') + 0$$

$$(1+x)^2 \frac{(y' \cos y)(1+x) - (\sin y)(1)}{(1+x)^2} = 16y^7 \cdot y' (1+x)^2$$

$$(1+x)y' \cos y - \sin y = 16y^7 y' (1+x)^2$$

$$-\sin y = 16y^7 y' (1+x)^2 - (1+x)y' \cos y$$

$$-\sin y = y' [16y^7 (1+x)^2 - (1+x) \cos y]$$

$$\frac{-\sin y}{16y^7 (1+x)^2 - (1+x) \cos y} = \frac{y' [16y^7 (1+x)^2 - (1+x) \cos y]}{16y^7 (1+x)^2 - (1+x) \cos y}$$

$$\boxed{\frac{-\sin y}{16y^7 (1+x)^2 - (1+x) \cos y}} = y'$$